Project Plan: Potential Energy Curve of H2 with DFT and a Quantum Algorithm

Oct 12, 2026

Team details

Team name	GroundStateDuo
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Project title	Potential Energy Curve of H2 with DFT and a Quantum Algorithm
Forked repository	github.com/tadeoye137/qff26xUIQC

Objective

We will compute the ground-state energy of the H_2 molecule as a function of the H–H distance with density functional theory (DFT) and with a quantum algorithm, and compare where the two agree and where they differ.

For each method we will extract the equilibrium bond length and the vibration frequency from the energy curve, and we will examine the stretched-bond region, where DFT is known to struggle.

Motivation

The University of Illinois materials science course now teaches this exact problem: place two protons at a given distance, compute the total energy point by point to get a potential energy surface, and solve it with DFT, with molecular dynamics, and on a quantum computer, then compare the three.

- **Why this problem:** it is small enough to finish in a week, and the result can be checked against the known H_2 bond length of about 0.74 Å.
- **Why the comparison matters:** DFT has trouble with strongly correlated systems, and quantum algorithms are being studied as a way to handle them. Stretched H_2 is the simplest case where that difference shows up.

- **Why our team can do it:** the DFT side uses Quantum ESPRESSO, which one member already uses for research, and the quantum side uses the Qiskit methods taught in the workshop.

Methods

All methods use the same list of about 12 H–H distances, from 0.5 Å to 3.0 Å, so the curves can be compared point by point.

1. **Exact diagonalisation (Qiskit).** Build the H_2 Hamiltonian at each distance and diagonalise it exactly. This is the reference curve.
2. **VQE (Qiskit, simulator).** Find the ground state with the variational quantum eigensolver using a chemistry-inspired ansatz and a classical optimiser. Compare against the exact curve to measure VQE accuracy.
3. **DFT (Quantum ESPRESSO).** Place the molecule in a large cubic box, use the Γ point only, and run `pw.x` at each distance with LDA or PBE. Repeat with spin polarisation for the stretched region, and run an isolated H atom in the same box to get the binding energy.
4. **Molecular dynamics (stretch goal).** If time allows, fit a Morse potential to the DFT curve and run a short dynamics simulation of the two atoms to obtain the vibration frequency.

Comparison. The methods use different basis sets, so absolute energies are not compared directly. Each curve is shifted to its own minimum, and the methods are compared by equilibrium bond length, vibration frequency, binding energy, and the shape of the curve at large distance.

Schedule

The plan is submitted before 14 October, and the work runs until 21 October.

Dates (October)	Task	Output
14–15	Set up Qiskit and Quantum ESPRESSO; run H_2 at the equilibrium distance with all methods	One working energy per method
15–16	Exact diagonalisation and VQE scan over all distances	Exact and VQE energy curves
17	DFT scan (unpolarised and spin-polarised) and isolated H atom	DFT energy curves and binding energy
18	Compare: shift curves to their minima, extract bond	Comparison plot and table

Dates (October)	Task	Output
	length and frequency	
19	Stretch goals: Morse fit and short MD, or VQE with a noise model	Vibration frequency from MD, or noisy-VQE curve
19–20	Write the report and prepare the presentation	Final report

Deliverables and success criteria

The project is complete when we can show one figure and one table that compare the methods.

- **Figure:** energy against H–H distance for exact diagonalisation, VQE, and DFT, each shifted to its own minimum.
- **Table:** equilibrium bond length, vibration frequency and binding energy for each method.
- **Report:** a short write-up explaining where the methods agree, where they differ, and why.
- **Success test:** VQE reproduces the exact curve closely, and the DFT bond length is close to 0.74 Å. A clear disagreement at large distance is also a valid result, if we can explain it.

Risks, fallbacks and reference

- **Simulator only:** all quantum results use a simulator, so no hardware queue time is needed. If real hardware is used, the backend name and date will be recorded, because IBM retires older machines.
- **VQE does not converge:** fall back to a smaller ansatz or a different optimiser, and report the exact-diagonalisation curve as the quantum-side reference.
- **DFT setup problems:** the hydrogen pseudopotential has to be copied to the cluster; if it is missing, run the DFT scan with PySCF, which is already installed with Qiskit Nature.
- **Molecular dynamics too much:** it is a stretch goal and can be dropped without changing the main comparison.

Reference. IBM-Illinois Discovery Accelerator Institute, "Quantum computing comes to materials science, in research and in class", 29 September 2026.